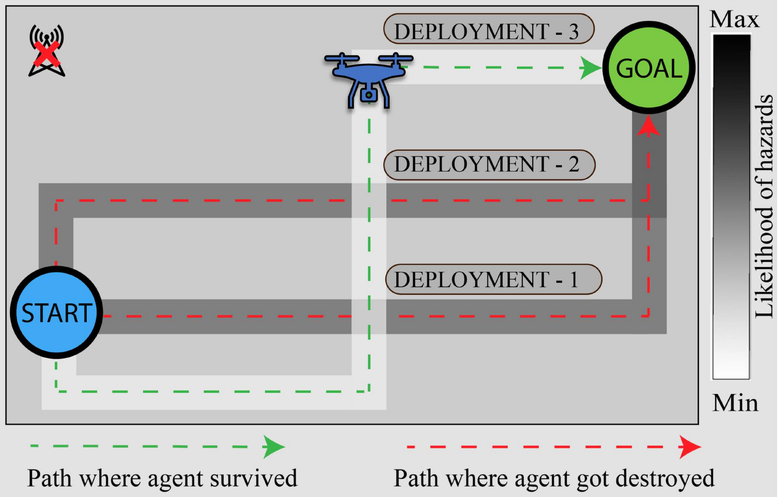
Estimating Hazardous Locations in Communication-Denied Environments: A Bayesian Network Approach with Path-Based Sensors

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MOTIVATION



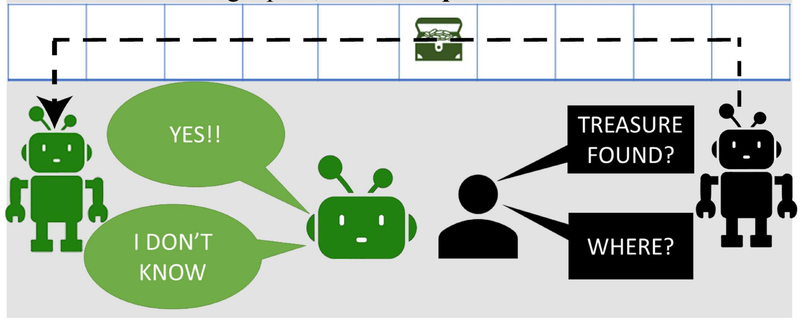
- In a communication-deprived environment, a hazard may ***detain***, ***disable***, or ***destroy*** the robot
- Such situations are common in disaster-stricken ***search & rescue missions, military operations*** in hostile environments, and ***under -water exploration*** and ***surveying***
- It is essential to ***gather as much information as possible*** in each deployment to avoid ***loss of robots***

RESEARCH OBJECTIVE

- Given a communication-denied environment with static hazards ***Z***, develop an ***efficient method*** to ***gather information*** about ***Z***
- To develop an ***exact inference mechanism*** based on a ***Bayesian network*** to infer from ***path-based sensor*** realizations

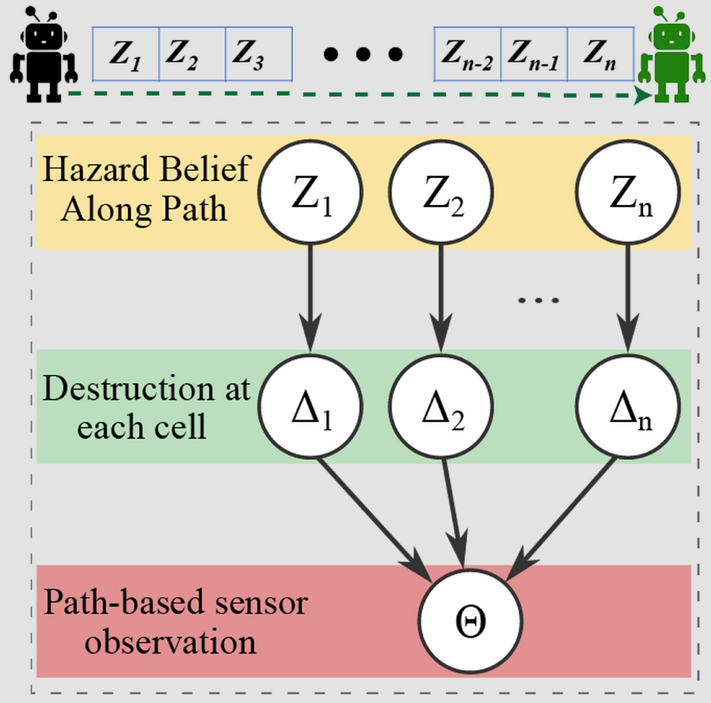
PATH-BASED SENSOR

A sensor that produces a ***binary observation*** indicating whether an event occured along a path, but not its ***precise location***.

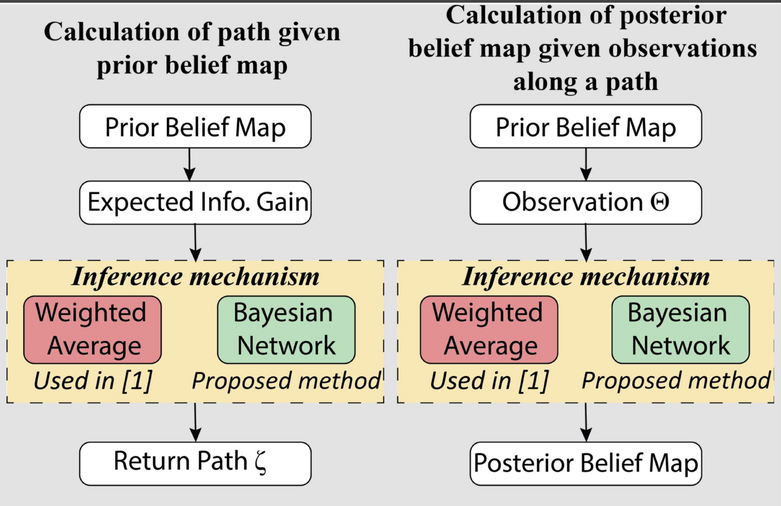


BAYESIAN NETWORK MODEL

If an agent traverses along the path $\zeta = \{ Z_1, Z_2, Z_3, \ldots Z_n \}$ and observes a path-based sensor trigger Θ, we can model this observation as a Bayesian network if we know the likelihood of destruction Δ_i for the ith cell.



PROPOSED STRUCTURE

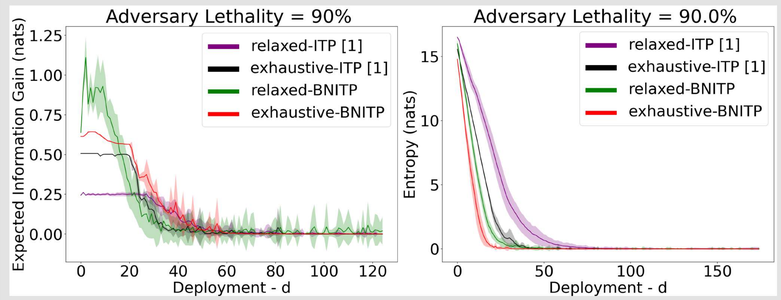


RESULTS

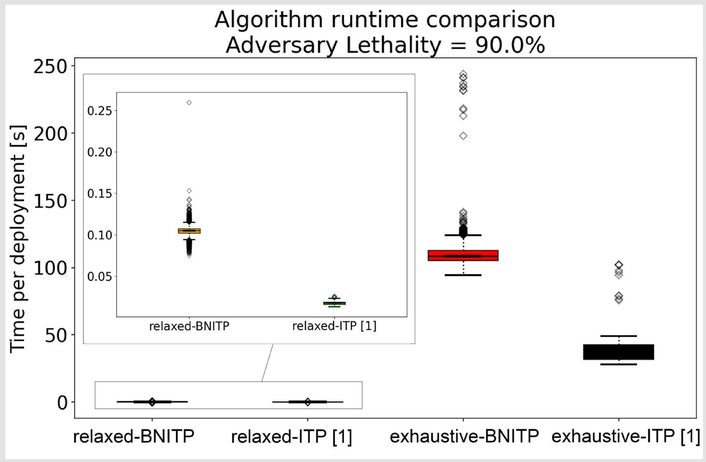
Two algorithms are proposed:

(1) **exhaustive-BNITP** (2) **relaxed-BNITP**

- **relaxed-BNITP** is a computationally feasible alternative to the **exhaustive-BNITP**
- Proposed **relaxed-BNITP** and **exhaustive-BNITP** outperforms [1] in terms of ***information gathering***



- **relaxed-BNITP** outperforms **exhaustive-ITP [1]** while taking approximately ***three orders of magnitude lesser time***



REFERENCES

[1] Otte, Michael, and Donald Sofge. "Path-based sensors: Paths as Sensors, Bayesian updates, and Shannon information gathering." IEEE Transactions on Automation Science and Engineering 18.3 (2021): 946-967.

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[3] Srivastava, Alkesh K., et al. "Bayesian Network Approach for Information Gathering Using Path-Based sensors" in-submission (2023)

[4] Srivastava, Alkesh K., et al. "Path-Based Sensors: Will the Knowledge of Correlation in Random Variables Accelerate Information Gathering?." arXiv preprint arXiv:2305.06929 (2023).